

Class 2

GDP, Living Standards, and Welfare

Part I — The Growth Miracle

The American Economy — draft course materials (proposal under discussion)

The point of today's class

Economic growth is one of the most important facts in human history — and while GDP is not the same as welfare, real income growth is tightly connected to almost everything people care about.

Why it matters

- ▶ Would you rather be a king in 1600 or an ordinary American today? The king had servants; you have antibiotics, anesthesia, and a supermarket.
- ▶ Make it concrete — an American home in 1870:
 - ▶ no plumbing, no electric light
 - ▶ laundry was a full day of physical labor
 - ▶ roughly half the household budget went to food
- ▶ For almost all of human history, ordinary people were poor, sick, and short-lived. Then, in a handful of places, everything changed — and the United States became the leading example.

Today

1. What GDP measures — and how to read it
2. The most important chart in economics
3. What growth buys: survival itself
4. What GDP misses
5. Does money buy happiness?
6. Welfare beyond GDP

What GDP measures

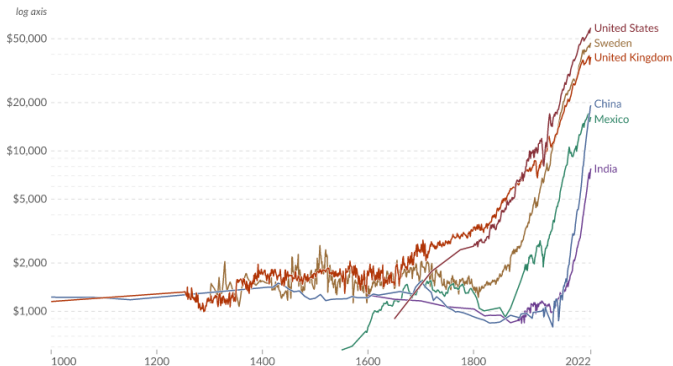
- ▶ $Y = C + I + G + NX$: everything produced and sold — consumption, investment, government purchases, net exports.
- ▶ **Real vs. nominal**: correct money for prices, or growth is an illusion of inflation.
- ▶ **Per person**: a bigger country is not a richer person.
- ▶ **Mean vs. median**: the average can hide the typical household.

GDP per capita since 1500

GDP per capita, 1000 to 2022

Our World
in Data

GDP per capita is a country's gross domestic product divided by its population. This data is adjusted for inflation and differences in living costs between countries.



Data source: Bolt and van Zanden - Maddison Project Database 2023

OurWorldinData.org/economic-growth | CC BY

Note: This data is expressed in international-\$ at 2011 prices.

Source: Maddison Project Database 2023 (Bolt and van Zanden), via Our World in Data. CC-BY 4.0.

Why small growth differences become enormous

- ▶ Growth **compounds**. Over 50 years:

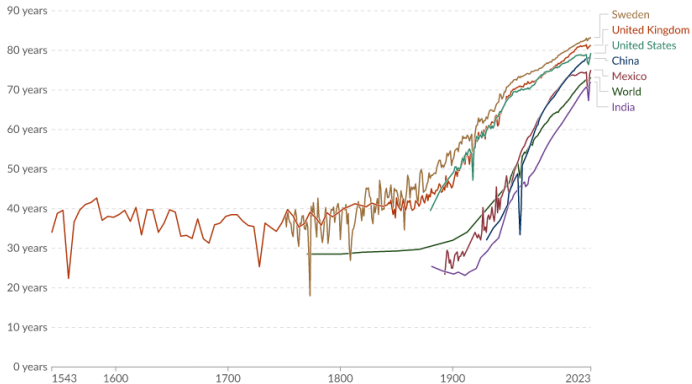
Annual growth	Income after 50 years
1% per year	1.6× today's
2% per year	2.7× today's
3% per year	4.4× today's

- ▶ Seemingly small differences in growth rates dominate almost everything else in the long run.

Life expectancy at birth, 1543–present

Life expectancy

Period life expectancy is the number of years the average person born in a certain year would live if they experienced the same chances of dying at each age as people did that year.



Data source: Riley (2005); Zijdeman et al. (2015); HMD (2025); UN WPP (2024)

OurWorldinData.org/life-expectancy | CC BY

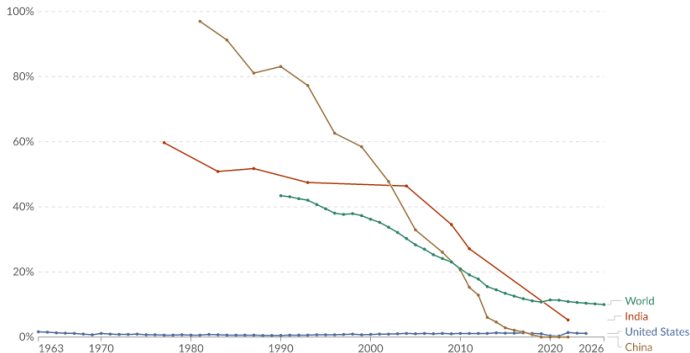
Source: Our World in Data, based on Riley (2005), Clio Infra, and UN WPP. CC-BY 4.0.

Extreme poverty has collapsed worldwide

Share of population living in extreme poverty, 1963 to 2026



Extreme poverty is defined as living below the International Poverty Line of \$3 per day. This data is adjusted for inflation and differences in living costs between countries.



Data source: World Bank Poverty and Inequality Platform (2026)

OurWorldinData.org/poverty | CC BY

Note: This data is expressed in international-\$ at 2021 prices. Depending on the country and year, it relates to income (measured after taxes and benefits) or to consumption, per capita.

Source: World Bank Poverty and Inequality Platform, via Our World in Data. CC-BY 4.0.

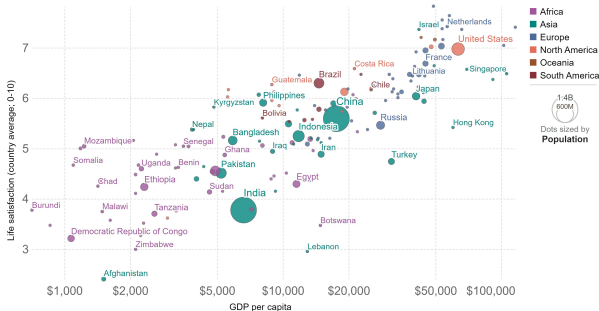
What GDP misses

- ▶ **Leisure:** working fewer hours makes you better off, not poorer — but GDP falls.
- ▶ **Home production:** a parent caring for a child adds nothing to GDP; paid daycare does.
- ▶ **Health and longevity:** living longer is worth a great deal and is not in GDP.
- ▶ **Distribution:** GDP per person says nothing about who gets what.
- ▶ **The environment:** pollution is not subtracted.

Does income buy happiness? Stevenson and Wolfers

Self-reported life satisfaction vs. GDP per capita, 2022

Self-reported life satisfaction is measured on a scale ranging from 0-10, where 10 is the highest possible life satisfaction. GDP per capita is adjusted for inflation and differences in the cost of living between countries.



Source: World Happiness Report (2023), Data compiled from multiple sources by World Bank

Note: GDP per capita is expressed in international-\$¹ at 2017 prices.

OurWorldInData.org/happiness-and-life-satisfaction/ • CC BY

1. **International dollars:** International dollars are a hypothetical currency that is used to make meaningful comparisons of monetary indicators of living standards. Figures expressed in international dollars are adjusted for inflation within countries over time, and for differences in the cost of living between countries. The goal of such adjustments is to provide a unit whose purchasing power is held fixed over time and across countries, such that one international dollar can buy the same quantity and quality of goods and services no matter where or when it is spent. Read more in our article: What are Purchasing Power Parity adjustments and why do we need them?

Source: Stevenson and Wolfers (2008), Brookings Papers on Economic Activity.

What about above \$75,000? Killingsworth (2021)

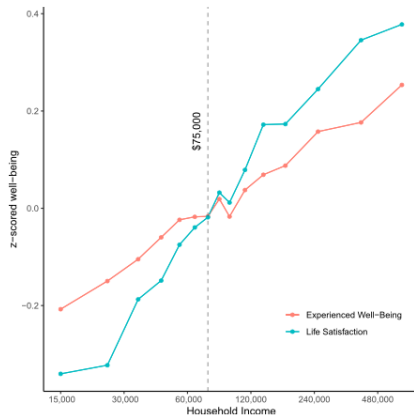


Fig. 1. Mean levels of experienced well-being (real-time feeling reports on a good-bad continuum) and evaluative well-being (overall life satisfaction) for each income band. Income axis is log transformed. Figure includes only data from people who completed both measures.

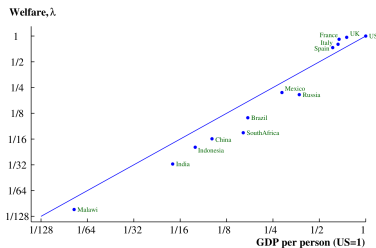
Source: Killingsworth (2021), PNAS, "Experienced well-being rises with income, even above \$75,000 per year."

Welfare vs. GDP across countries (Jones–Klenow)

WELFARE ACROSS COUNTRIES AND TIME

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Figure 5: Welfare and Income across Countries



(a) Welfare and income are highly correlated at 0.98...

The ratio of Welfare to Income



Source: Jones and Klenow (2016), *American Economic Review*, "Beyond GDP? Welfare across Countries and Time."

Reading a chart critically — the habit of the course

Every empirical class asks the same three questions:

1. What is the **comparison**?
2. What is the **counterfactual**?
3. How could this chart **mislead** us? (log scales, cherry-picked windows, mean vs. median, correlation vs. causation)

Discussion

France looks much poorer than the U.S. in GDP per person — but the French work fewer hours and live longer. Which country is actually better off? What would you need to measure to answer?

Read before next class

- ▶ Deaton, *The Great Escape* (excerpt) — accessible anchor.
- ▶ Stevenson and Wolfers (2008), “Economic Growth and Subjective Well-Being” (skim the figures).